Exercise 12: Pulley

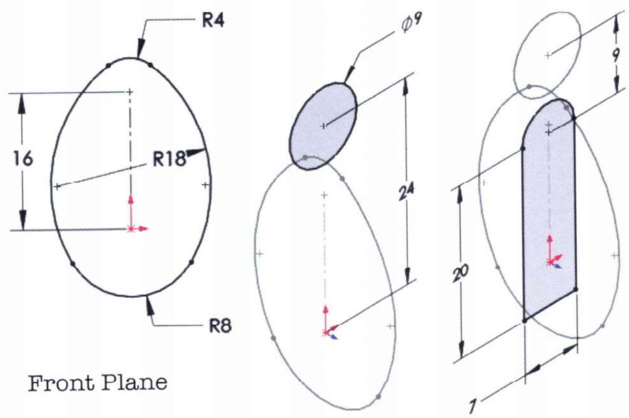
Complete this part using the dimensions provided. Use relations where applicable to maintain the design intent.

This lab uses the following skills:

- *Symmetry in the Sketch* on page 121
- *Mid Plane Extrusion* on page 123
- *Draft Toggle* on page 123
- *Introducing: Circles* on page 126

Optional Sketching

If you would like to use the existing geometry, skip to **Procedure**. If you would like to create the sketches, open a new **mm** part and use the dimensions below. There are three sketches required, the first created on the Front Plane, the remaining two on the Right Plane.



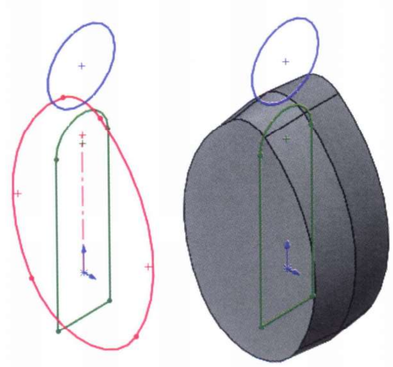
Front Plane

Right Plane

Procedure

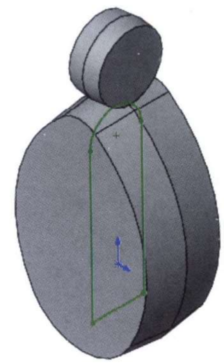
Open an existing part named Pulley.

1 **Extrusion with draft.**
Extrude the Base sketch (red) **10mm** using the **Mid-plane** end condition and **6°** of draft.



2 Hanger.

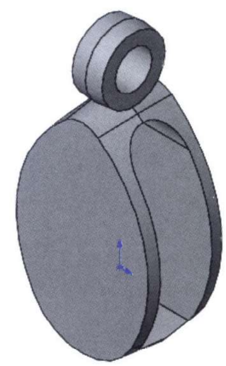
Use the Hanger (blue) sketch and another **Mid-plane** extrusion of **4mm** with the same amount of draft.



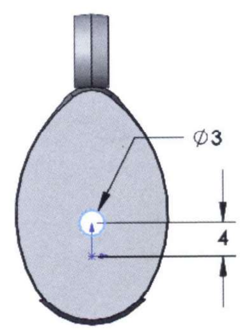
3 Cut and hole.

Create a cut using the Center Cut sketch (green). The cut is **Through All- Both**.

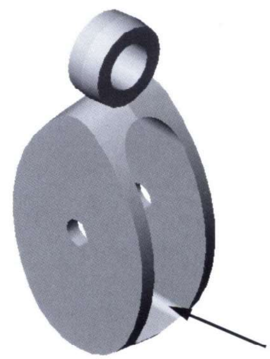
Add a **5mm** diameter hole.



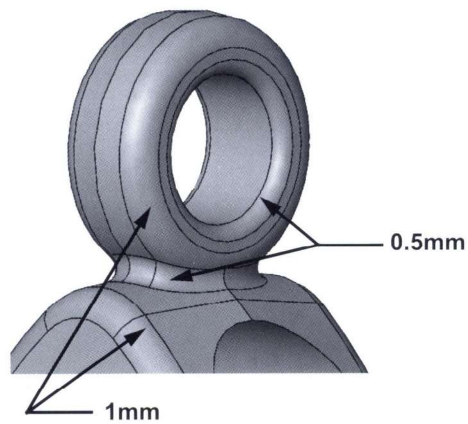
Create another hole, **3mm**, centered above the origin.



4 Fillets.
Add a fillet of **1mm** to the bottom edges of the cut.



Add fillets of **1mm** and **0.5mm** as shown.

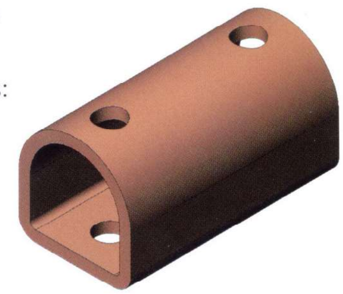


Note These fillets are very order dependent; the **1mm** fillets must precede the **0.5mm** ones.

5 Save and close the part.

Exercise 13: Symmetry and Offsets 1

Use offsets and symmetry to complete the part.



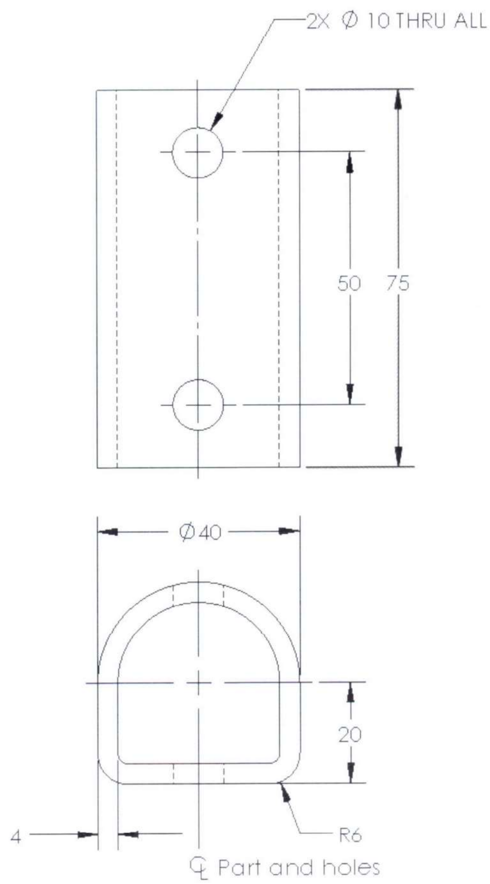
This lab reinforces the following skills:

- *Introducing: Centerline* on page 121
- *Symmetry after Sketching* on page 122
- *Mid Plane Extrusion* on page 123
- *Sketching an Offset* on page 137

Units: **millimeters**

Dimensioned Views

Use the following graphics to create the part.

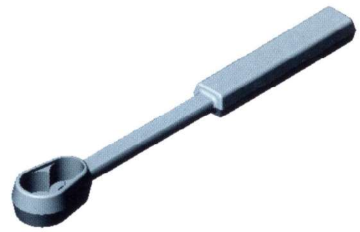


Exercise 14: Ratchet Handle Changes

Make changes to the part created in the previous lesson.

This exercise uses the following skills:

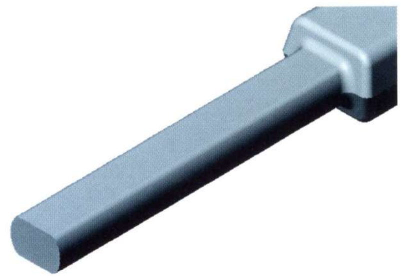
- *Trim and Extend* on page 139



Design Intent

Some aspects of the design intent for this part are:

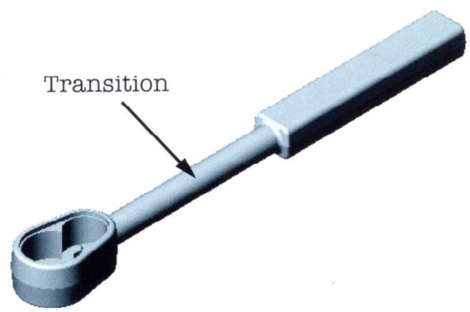
1. The part must remain symmetrical about the Right plane.
2. The Transition requires flats that are driven by the distance between them.



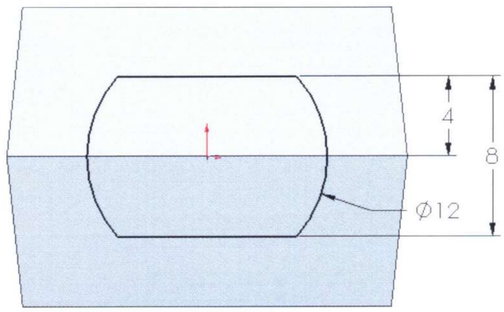
Procedure

Open an existing part.

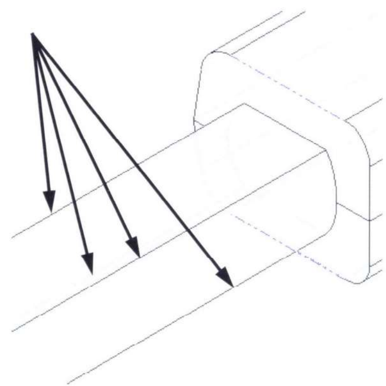
1 **Open the part Ratchet Handle Changes.**
The change will take place in the shape of the Transition feature.



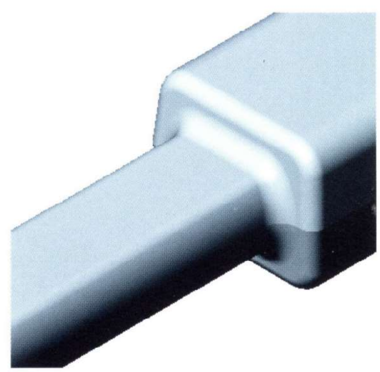
2 Edit the sketch.
Right-click the `Transition` feature and click **Edit Sketch**. Modify the sketch to add the equally spaced horizontal flats **8mm** apart. Exit the sketch.



3 Edit Feature.
Edit the `H End Fillets` feature to add more edges. Select the four new edges created by the flats. Click **OK**.



4 Resulting fillets.
The new edges become part of the fillet feature, causing the shape of the next fillet feature to update.



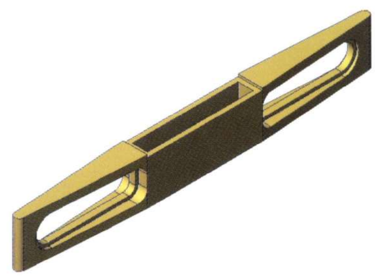
5 Save and close the part.

Exercise 15: Symmetry and Offsets 2

Use offsets and symmetry to complete the part.

This lab reinforces the following skills:

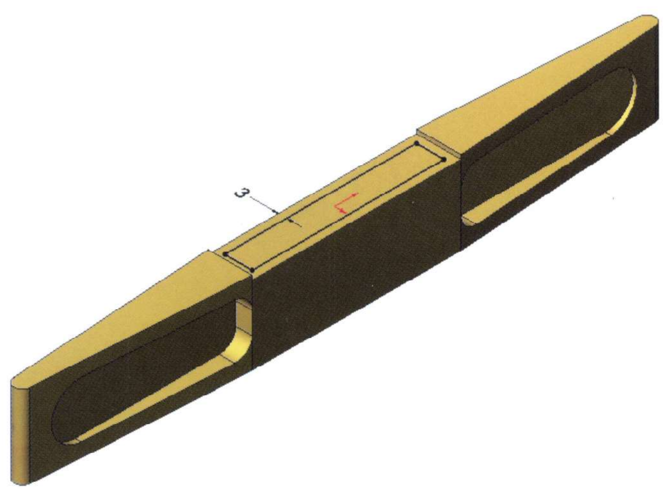
- *Introducing: Centerline* on page 121
- *Symmetry after Sketching* on page 122
- *Introducing: View Normal To* on page 126
- *Sketching an Offset* on page 137
- *Offset From Surface* on page 142
- *Measuring* on page 144



Procedure

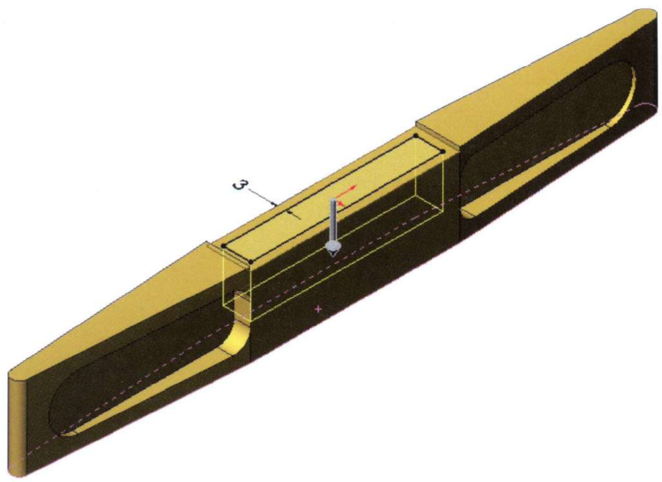
Open the existing part `Offset_Entities` and create the geometry as shown in the following steps.

1 **Offset entities.**
Use **Offset Entities** to create the sketch geometry shown.

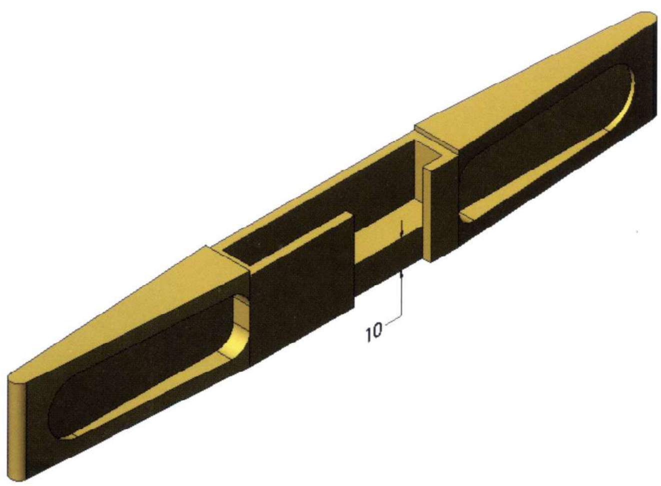


2 Offset from surface.

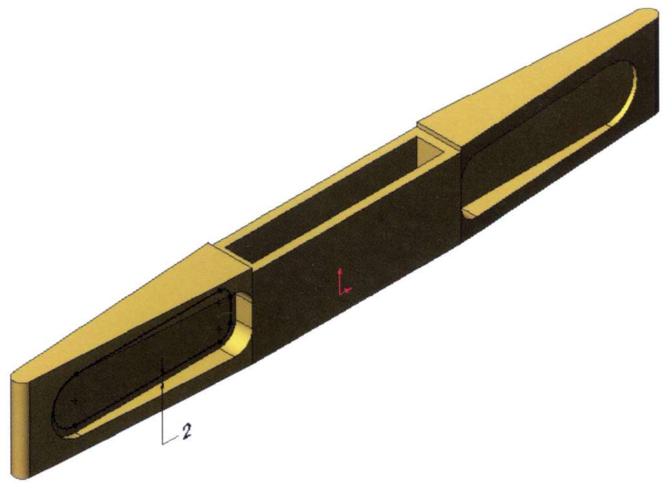
Use a **Cut Extrude** with an **Offset From Surface** of **10mm.** Select the hidden face using **Select Other**.



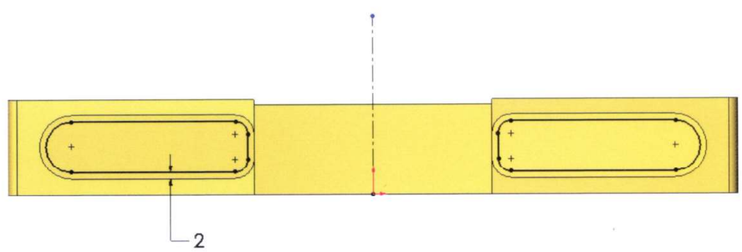
This cutaway shows the results of the offset from surface option. The **Measure** tool can be used to check this value.



3 Offset.
Create a new sketch and offset the face **2mm**.



4 Offsets with symmetry.
Use **View Normal To** to orient the part. Create the geometry using **Offset Entities**. Use a **Cut Extrude Through All**.



5 Save and close the part.

Exercise 16: Tool Holder

This lab reinforces the following skills:

- *Symmetry in the Sketch* on page 121
- *Mid Plane Extrusion* on page 123
- *Introducing: Circles* on page 126
- *Trim and Extend* on page 139

Units: **millimeters**

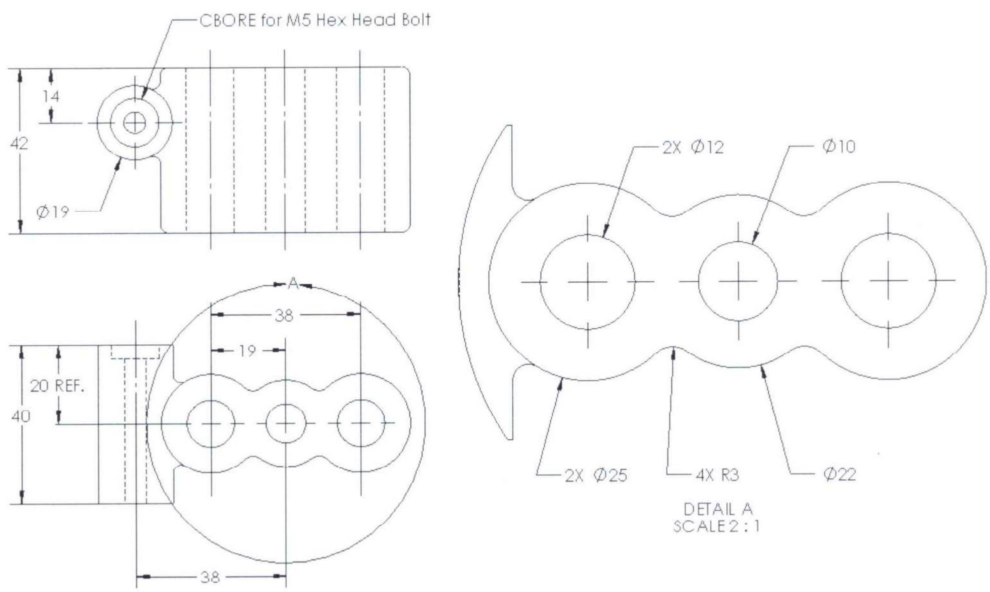
Design Intent

Some aspects of the design intent for this part are:

1. All fillets and rounds are **R2mm** unless otherwise noted.
2. Circular edges of equal radii/diameter should remain equal.

Dimensioned Views

Use the following graphics with the design intent to create the part. Tangent edges are hidden in all views.



Exercise 17: Idler Arm

Create this part using the dimensions provided. Use relations where applicable to maintain the design intent. Give careful thought to the best location for the origin.

The only reference planes required for this part are Top, Front and Right.

This lab uses the following skills:

- *Symmetry in the Sketch* on page 121
- *Mid Plane Extrusion* on page 123
- *Extruding Up To Next* on page 128

Units: **millimeters**

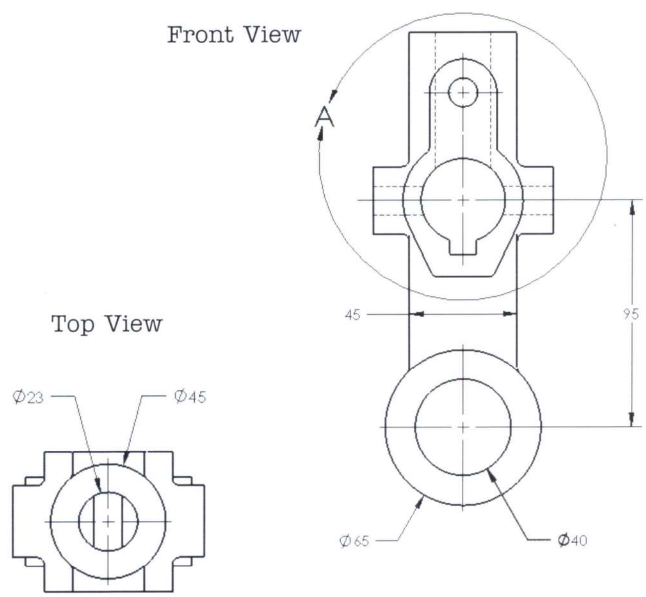
Design Intent

The design intent for this part is as follows:

1. The part is symmetrical.
2. Front holes are on the centerline.
3. All fillets and rounds (highlighted red above) are **R3mm**.
4. Center holes in Front and Right are in line.

Dimensioned Views

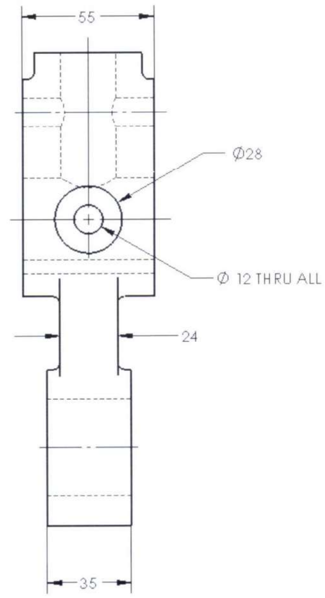
Use the following graphics with the design intent to create the part.



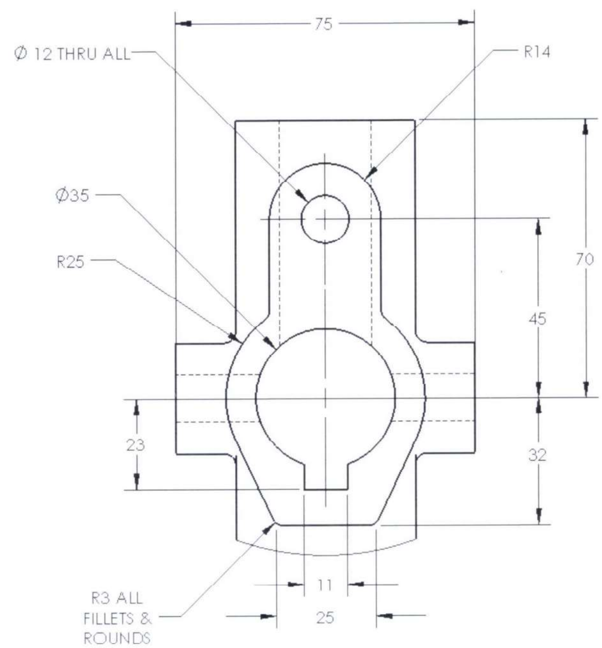
Exercise 17
Idler Arm

SOLIDWORKS

Right View



Detail View A



DETAIL A
SCALE 1 : 1

Exercise 18: Up To Surface

Use symmetry and an up to surface end condition to complete the part.

This lab reinforces the following skills:

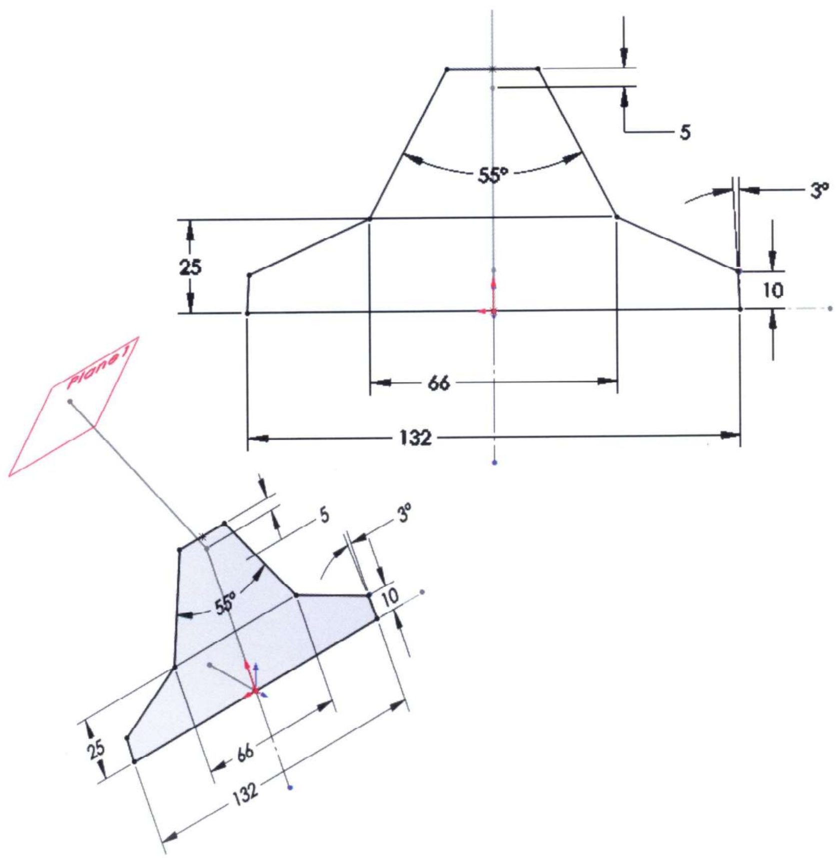
- *Introducing: Mirror Entities* on page 122
- *Up To Next vs. Up To Surface* on page 128

Units: **millimeters**

Procedure

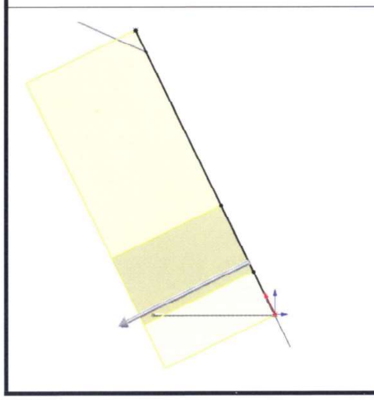
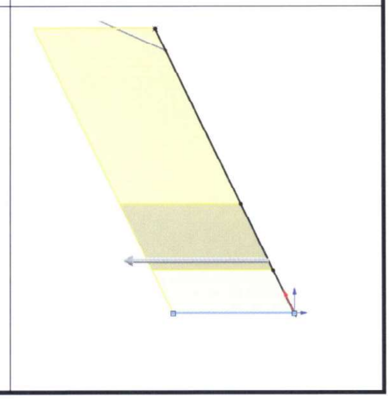
Open an existing part named `Up_To_Surface`.

1 **Sketch.**
Using `Plane2`, create the geometry below using lines and symmetry.

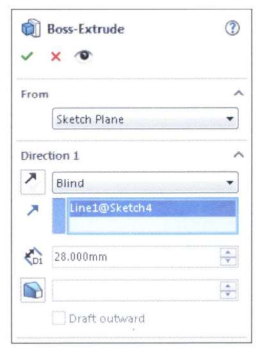


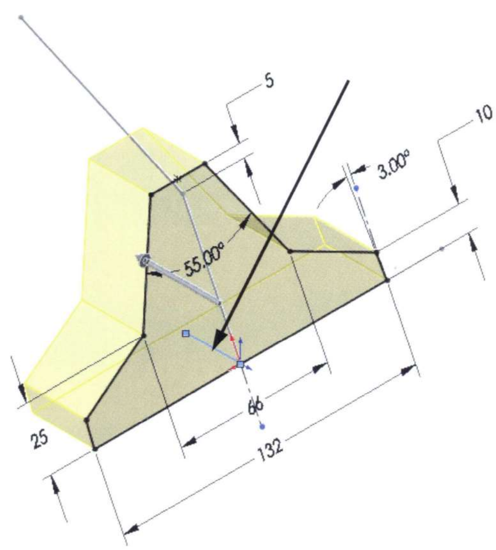
Using Direction of Extrusion

The default extrusion direction is normal to the profile or sketch plane. Occasionally a model dictates an extrusion that is not normal to the profile, but along a vector using the **Direction of Extrusion** option. The vector can be a sketch line, edge, or axis.

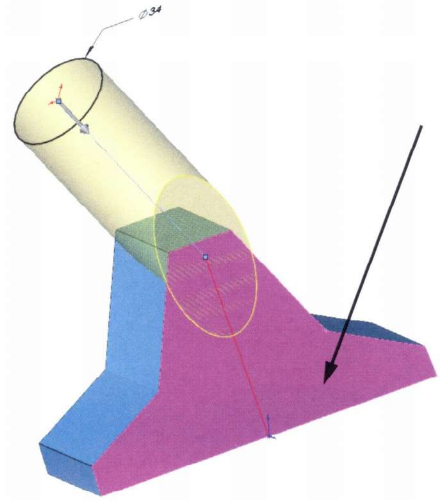
Default Extrusion	Direction of Extrusion (line)
	

2 Extrude with direction.
Extrude the sketch. Click in the **Direction of Extrusion** field and click the blue sketch line as shown. Set the **Depth** to **28mm**.

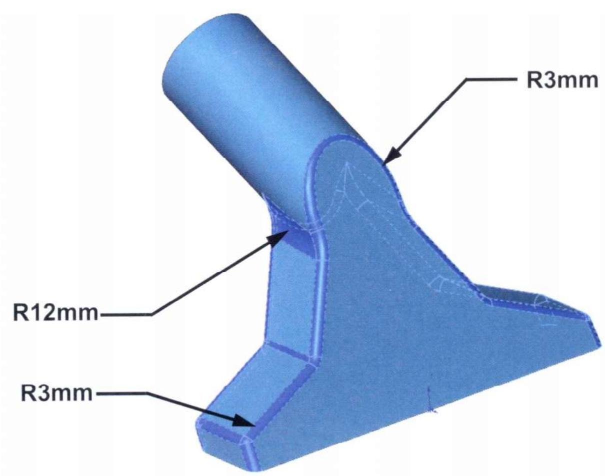




3 Up to surface.
Sketch on `Plane1` and create the circle dimensioned **D34mm** as shown. Extrude the sketch **Up To Surface** and click the pink face to terminate the extrusion.



4 Fillets.
Add fillets as shown below.



Tip

These fillets are very order dependent; the **12mm** fillets must precede the **3mm** ones.

5 Save and close the part.